Comparing the effectiveness of variational autoencoders and autoencoders at anomaly detection.

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1.0 Executive Summary

The primary objective of this project is to compare the anomaly detection capabilities of an autoencoder and a variational autoencoder using time series data. Autoencoder and a variational autoencoder have similar structure. Both techniques are a combination of encoder and a decoder. However, they are fundamentally different. An autoencoder encodes the input to a vector whereas variational autoencoder encodes it to a distribution.

The secondary objective of this project is to evaluate the effectiveness of these models with time-series data. Autoencoder is listed as an anomaly detection algorithm in a comprehensive study of timeseries anomaly detection[1]. Autoencoder was listed as a semi-supervised algorithm dealing with multivariate timeseries data. Autoencoders have no inherent mechanism like RNN that can be useful for capturing relation between previous or next data points. Despite this autoencoders are widely used for anomaly detection in time series data. This project will test the anomaly detection capabilities of an autoencoder and a variational autoencoder with timeseries data.

Both the algorithms are used to build models where they are trained with data with no anomalies and tested with data with anomalies. These models are called semi supervised models. The models are made comparable by keeping the structure as similar as possible. This was done relatively easily because models have a similar encoder-decoder architecture.

The timeseries dataset chosen for the evaluation of the models is a modified version of PhysioNet/CinC Challenge[2]. The dataset used for our evaluation is a subset of the original dataset. The dataset was modified to suit the semi supervised nature of the models. The dataset is split into training and test set. The training set consists of data with no anomalies and the test set consists of data with 4 categories. The 4 categories are normal, atrial fibrillation, other rhythms and noisy data. The categories atrial fibrillation, other rhythms and noisy data are merged into anomalous data category. This gives us two categories normal and abnormal.

The results obtained by the models are evaluated using reconstruction error. The data with high reconstruction error is anomalous data and data with low reconstruction error is normal data. The new labels are compared to the old labels to obtain confusion matrix. The values obtained from the confusion matrix can be used for the evaluation of the data.

2.0 Introduction:

Autoencoders and variational autoencoders have similar encoder-decoder architecture. However, they are fundamentally different. An autoencoder used the encoder to encode the input into a vector and then decodes it using a decoder. A variational autoencoder uses an encoder to encode the input data into a distribution into a vector and then uses the decoder to decode it. This project aims to compare the anomaly detection capabilities of encoding the input to a vector versus encoding the input to a distribution. There are several algorithms derived from an autoencoder like LSTM based encoder-decoder[3] which can be used for anomaly detection. However, this study will employ a classic autoencoder for anomaly detection[4]. Similarly, a classic variational autoencoder for anomaly detection[5] structure is used. Both models are designed in a semi-supervised format. The models are trained using a data which is normal, and the model is tested using a mixture of normal and abnormal data.

The autoencoder is built using standard dense layers. The encoder and decoder have a symmetric architecture. The encoder for the Autoencoder has input layer feeding data to the dense layers. The dense layers feed the data an additional layer which holds the encoding. The encoder encodes the data into a vector. Therefore, encoding layer is a simple dense layer. The decoder receives the encoded data in a vector form and tries to reconstruct the data to its original form. This autoencoder setup uses a standard mean square error loss function to compare input to output and adjust the parameters.

The variational autoencoder has a similar structure where the input is passed to the input layer. The input is feed to the dense layers. These layers feed the data to 2 layers which are used to hold the information about the distribution. These layers are then connected to a layer which samples the distribution. This information is then passed to the set of dense layers acting as decoders which try to reconstruct the data to the original shape. The loss function used for this variational auto encoder is a sum of mean square error loss and the K L divergence loss. The additional loss function added, K L divergence loss function, helps the variational autoencoder learn the distribution.

The outputs of the models are obtained and separated based on reconstruction loss. The labelled output data is then compared with the original labels to obtain a confusion matrix. The evaluation criteria for the model are accuracy, precision with abnormal data and recall with abnormal data. These criteria will determine effectiveness of model at abnormal data detection.

Additionally, this project also assesses effectiveness of the standard encoder-decoder combination with time-series anomaly detection tasks. A comprehensive study of time series anomaly detection[1] methods has listed autoencoder as model for evaluation. Autoencoder does not have a system for capturing temporal information like long-short term memory model. In this project the results obtained by the encoder-decoder models are analysed and the performance of the model is evaluated. This is done by evaluating values like precession and recall for the abnormal values from a general perspective.

3.0 Literature Review:

Anomaly detection in time series data is heavily researched. There are several methods that can be used for anomaly detection. There are several studies done to evaluate the timeseries algorithms [1]. This project mainly focuses on the anomaly detection capabilities of 2 algorithms. Autoencoder and the variational autoencoder.

Autoencoder models are often used for anomaly detection[4]. The autoencoder in uses an encoder-decoder architecture to deconstruct and reconstruct a signal. Autoencoder stores the deconstructed signal in a vector. This algorithm is semi-supervised. The model is trained using normal data. When testing this model, a combination of normal and abnormal data is passed to the model. In the testing dataset the normal data is reconstructed closer to the original data passed into the model, the model would have trouble reconstructing the model to the original form. Reconstruction error is used to measure the error that is obtained during reconstruction. Formula for the reconstruction error is displayed below.

$$Err(i) = \sqrt{\sum_{j=1}^{D} (x_j(i) - \hat{x_j}(i))^2}$$

Normal data should have a small reconstruction error whereas the abnormal data have a very high reconstruction error. A threshold value is set based on the values of reconstruction error obtained. If the value of the reconstruction error is higher than the threshold the value is abnormal. If the value of reconstruction error is lower than the value than the threshold the value is normal.

The variation autoencoder encodes the inputs into a distribution. The objective of the variational autoencoder is to learn the distribution of the input data. There is sampler that samples the distribution, and the decoder decodes the sample. The data learns the distribution of the data using K L divergence loss (Kullback–Leibler divergence). This loss function helps the function learn distribution.

$$\frac{1}{2}\sum_{k}(e^{\Sigma(X)} + \mu^2(X) - 1 - \Sigma(X))$$

This loss is added to the standard mean square error loss function.

$$\sum_{i}^{N}(x_i - \hat{x}_i)^2$$

The combined loss function is used to learn the mean and the standard deviation of the data. The mean square error loss function is used to match the input to the output. Adding KL loss in addition to the mean square error loss will enable the data to learn the distribution of the model. The KL loss will compare the data the learned model distribution to a standard normal distribution.

The distribution is sampled by the sampler. The sampling is done using reparametrizing the mean and the standard deviation learned. Back propagation and sampling together are challenging. To sample the data intuitively while allowing the model to learn is done using reparameterization. There is a

additional value ε which the reparameterization parameter that is learned outside the network. This parameter ε is calculated using the mean (μ) and the standard deviation (σ) learned in the network. The formula 'z = $\mu + \varepsilon\sigma$' is used to get a random sample z.

The anomalies in a variational autoencoder are detected by comparing the inputs to the reconstructed outputs. The reconstruction loss is calculated and the data is labelled using determined threshold.

4.0 Methodology:

4.1 Data Pre-Processing:

The data used for this experiment is a modified version of PhysioNet/CinC Challenge[2] dataset. This dataset was modified to be suitable for a semi-supervised algorithm. The data used to train the model contains normal signals or recordings of normal sinus rhythm. The test data contains a mixture of 4 categories of signals from the original normal sinus rhythm, atrial fibrillation, other rhythms and noise. The task was to detect anomalies, therefore atrial fibrillation, other rhythms and noise are combined into a single category abnormal. The signals from these categories are randomly chosen from the original dataset. All signals are sampled at 300 hertz. There are signals of varying lengths from 9 seconds to 30 seconds.

Data in the train and test set must be pre-processed before it can used in model building. The data obtained has two major problems. Each signal has a different mean and standard deviation. This will be very problematic during training process this might make it difficult for the model to learn. All the signals have been standardized so that all the signals have mean of 0 and a standard deviation of 1. The second problem with dataset is the irregular size of each signal. The range of the length of signal is 3000 to 18000 readings. There are some signals that are going outside the range. Padding the signal with 0s would hinder with the timeseries nature of the data. Therefore, to retain the most information The signals are cropped and split into different to form multiple readings. All the signals are split into multiple signals with 3000 readings in each signal. For example, a signal with 9000 readings is split into 3 signals with 3000 readings each. The maximum length of the signal is little over 18000. Therefore, there are 6 marks placed for splits 3000, 6000, 9000, 12000, 15000 and 18000. When the signal is length is over a certain mark appropriate number of splits are made. The conditions to check if the signal is over a mark is checked from the highest mark to the lowest mark. When a signal has more than readings than a specified mark but less readings than the next mark the lower mark is considered. This means that only the readings over the mark are lost. This causes data loss from certain signals. There are a few signals with less than 3000 readings these signals are also eliminated. Despite the data loss, majority of the signal behaviour is captured for all signals.

A different pre-processing style is applied to the test data. Unlike train data the top part of test data is considered. The first 3000 readings for all signals are picked and the rest of the signal is discarded. The anomalies are present throughout the signals for anomalous data therefore the first half of the signal is enough to represent the anomaly.

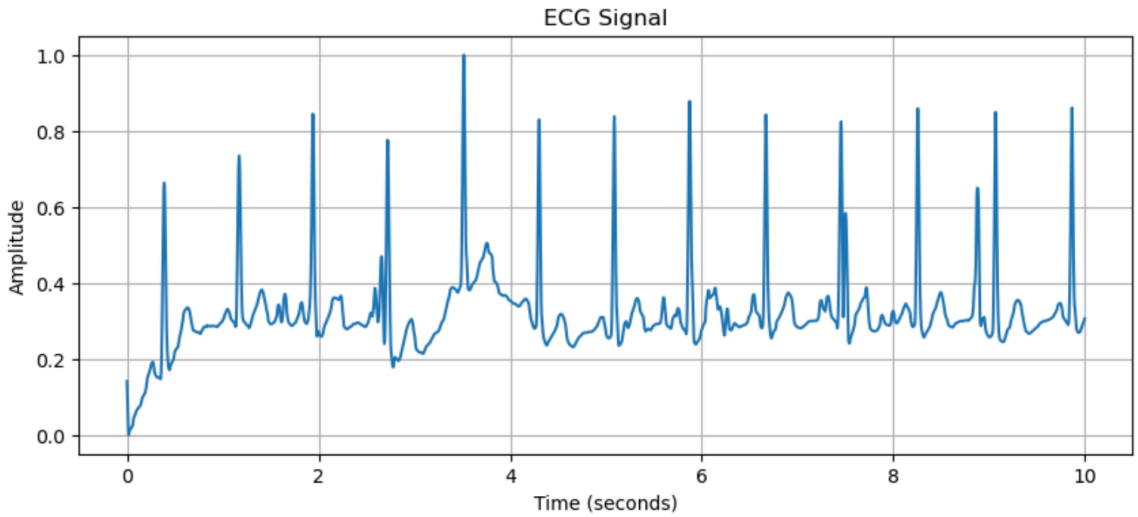

Figure 1

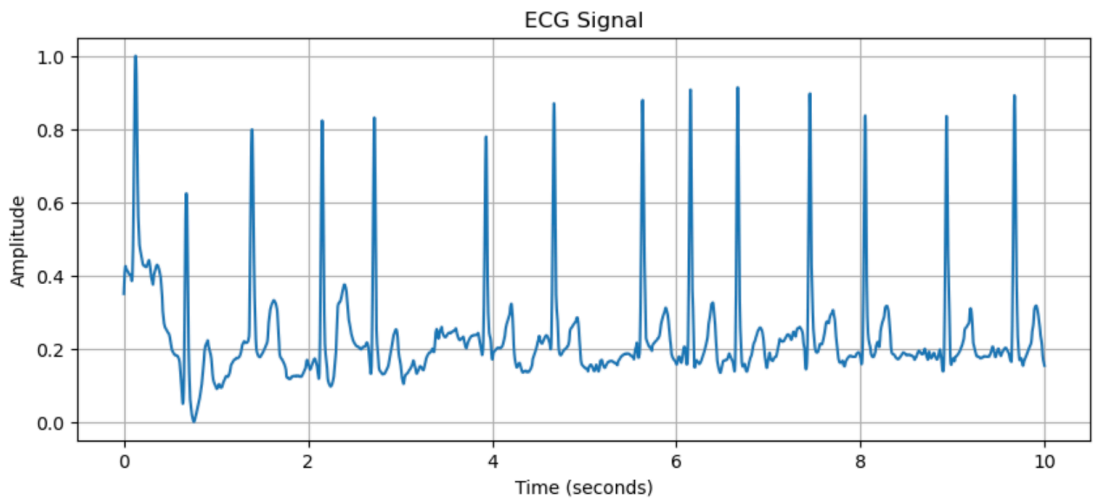

Figure 2

The readings plotted in figure 1 are from the train training set. The readings plotted in figure 2 are test set. The test set readings are taken from a signal categorized as abnormal more specifically the atrial fibrillation category.

4.2 Models:

For this project two models are built. The first model is the autoencoder model. It consists of 3 components encoder, decoder and an encoded vector. For this autoencoder mean square loss is used as a loss function. The structure of the model is kept relatively simple. The structure of the autoencoder is displayed below.

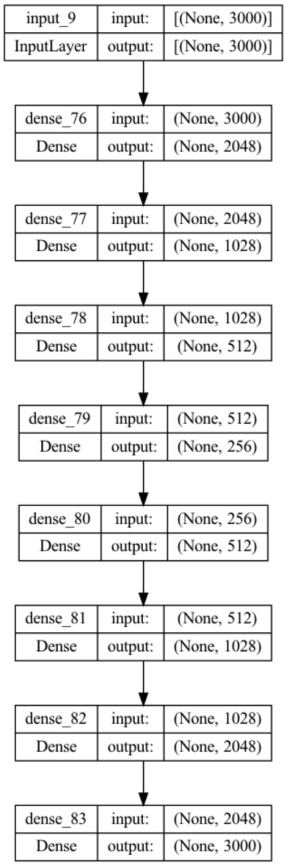

Figure 3

The autoencoder has an input layer to take the input. The input is passed through a sequence of 3 dense layers ‘dense_76’, ‘dense_77’ and ‘dense_78’. These layers together are the encoder. These layers learn to create encoding during the training. The encoding is stored in ‘dense_79’. As shown by the Figure 3 the size of the encoding is 256. ‘dense_80’, ‘dense_81’ and ‘dense_82’ are the decoder part of the decoder. ‘dense_83’ is also part of the decoder. This layer is used to convert the data to a suitable output size.

The other model used is the variational autoencoder. This model consists of 4 parts encoder, encoded distribution, sampler and the decoder. The decoder is also used to sample data from the captured distribution. The loss function used is KL divergence. The structure is similar to autoencoder structure to make the models comparable.

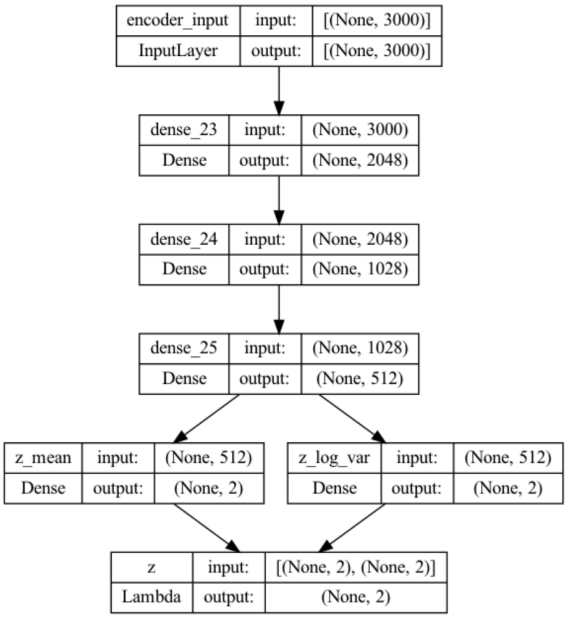

Figure 4

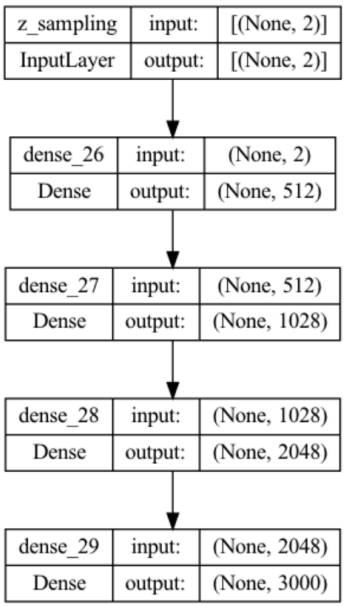

Figure 5

Figure 4 contains the encoder, the encoded distribution and the sampler. Figure 5 contains the decoder. Figure 4 and Figure 5 together show the structure of variational autoencoder model. The 'encoder_input' is the input layer that takes the input. 'dense_23', 'dense_24' and 'dense_25' are part of the encoder. These layers are trained to lean or capture the distribution. 'z_mean' and 'z_log_var' layers are used to capture the mean and standard deviation of the distribution the model is trying to learn. These layers are standard dense layers that are used to store the encoding. The 'z' is the sampling used for sampling the distribution. 'z' layer is lambda layer, and a special function is used to reparametrize the data and acquire a sample from the distribution. 'z_sampling', 'dense_26', 'dense_27' and 'dense_28' are part of the decoder. These decoders are used to reconstruct the original signal from the sampled distribution. 'dense_29' is also part of the decoder used to convert the data into suitable output size.

The models are trained using training data. Both models are trained with the same amount of data. The models are trained with the same batch size. These models are trained for the same number of epochs.

4.3 Testing and obtaining results:

The testing process for both models are similar. Both models are tested using the same testing data created during data pre-processing. The test data is inserted into the model. The output is the reconstructed signal. The reconstruction error is calculated for all signals. A threshold for the reconstruction error is obtained based on to the distribution below.

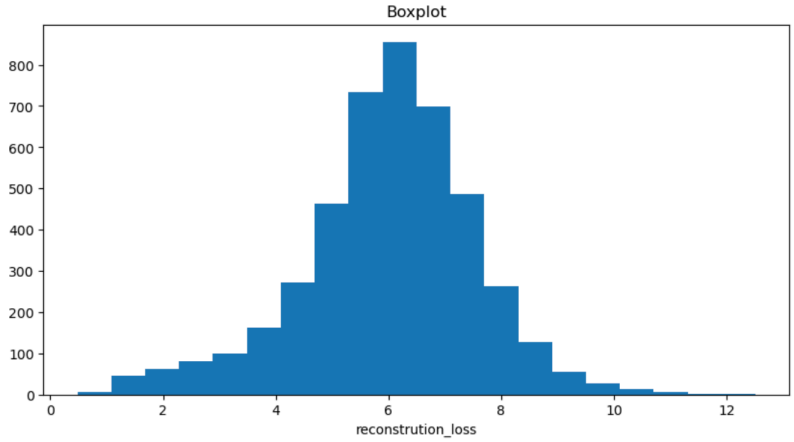

Figure 6

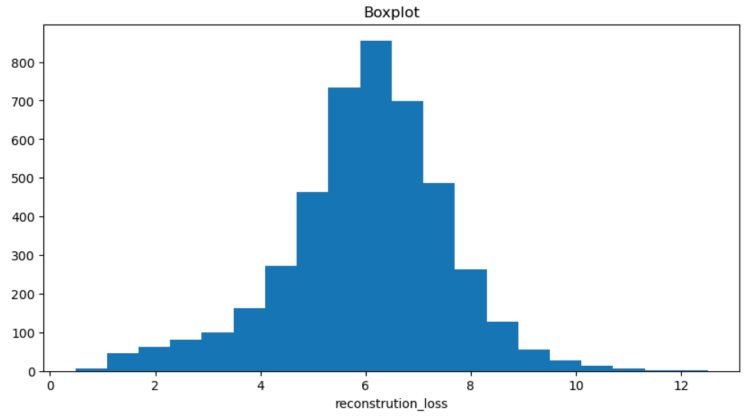

Figure 7

Figure 6 shows a histogram of the reconstruction loss of the results obtained from the autoencoder. Figure 8 shows a histogram of the reconstruction loss of the results obtained from the variational autoencoder.

Based on the graphs shown in figure 6 and figure 7 reconstruction error is decided to 6. Using this threshold also produces results that have with a similar number of normal and abnormal readings as the test data produced. The labels are assigned to the test data. These labels are compared to the original labels.

5.0 Findings:

The predicted labels are compared to the actual labels and a confusion matrix is constructed.

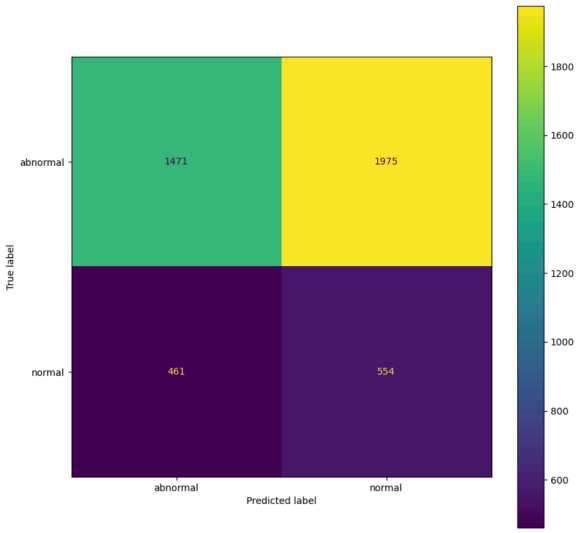

Figure 8

Figure 8 shows the confusion matrix obtained by comparing the results of an autoencoder to true labels. From this a lot of values like F1 Score, precision and recall can be obtained. The accuracy of the auto encoder is 45.39 precent.

Table 1

Autoencoder	Normal	Abnormal
Precision	0.2191	0.7614
Recall	0.5458	0.4269
F1 score	0.3127	0.5564

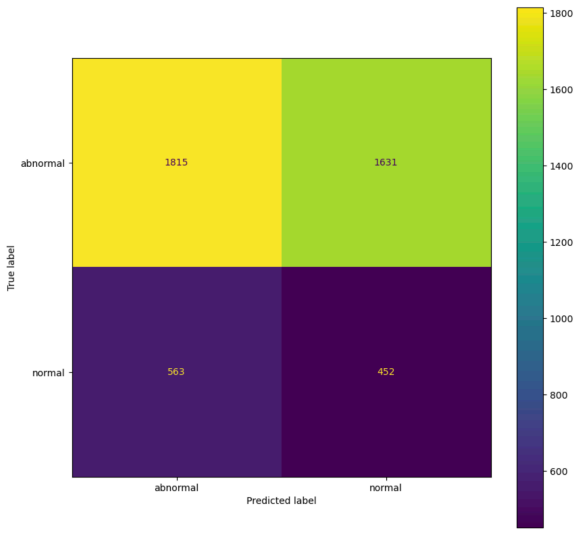

Figure 9

Figure 9 shows obtained by comparing the results of a variational autoencoder to true labels. From this a lot of values like F1 Score, precision and recall can be obtained. The accuracy of the model is 50.82 percent.

Table 2

Variational Autoencoder	Normal	Abnormal
Precision	0.2169	0.7632
Recall	0.4453	0.5267
F1 score	0.2917	0.6233

Evaluating results:

This section analyses results obtained in table 1 and table 2. Results in table 1 are obtained by autoencoder model. Results in table 2 are obtained by variational autoencoder model.

The autoencoder model has an accuracy of 45.39 percent. This shows that the model is not quite accurate. The autoencoder has a precision of 76.14 percent. The readings that were classified as abnormal are right 76.14 percent of the time. The readings that were classified as normal are right only 21.91 percent of the time. The model classified several abnormal signals as normal. The model only classified 42.69 percent of the abnormal data and 54.58 percent of the normal data correctly. This shows that the autoencoder is struggling with detecting abnormal readings.

The variational autoencoder model has an over accuracy of 50.82 percent. Variational autoencoder appears to be more accurate. The variational autoencoder has a precision of 76.32 percent. The readings that were classified as abnormal were right 76.32 percent of the time. The readings that were classified as normal are right only 21.69 percent of the time. This model is also classified several abnormal signals as normal. The model only classified 52.67 percent of the abnormal data and 44.53 percent of the normal data correctly. Variational autoencoder is doing better at detecting abnormal data.

The recall shows the percentage of a category that has been detected by the algorithm. Precision shows the percentage to the detected readings that are classified correctly. Both models have about the same amount of precision for detecting abnormal data. This shows that both models have the same performance for classifying the detected abnormal data correctly. The precision of the both the model with normal data is low. This shows that a lot of signals are classified as normal but only a few of these signals belong to the normal category. The recall for abnormal data of a variational auto encoder is higher than the autoencoder. This shows that the variational autoencoder is better at detecting abnormal data. The autoencoder has a higher recall for normal data than variational autoencoder. This shows that the autoencoder is better at detecting normal data.

Conclusion:

Based on the results obtained from the experiment variational autoencoder has a higher chance of detecting anomalies or abnormal values than an autoencoder. The overall f1 score for the category show that the autoencoder has better performance with normal data and variational autoencoder has a higher performance with abnormal data. For strictly detecting anomalies variational autoencoder has higher performance than an autoencoder.

Neither of the models are good at detecting anomalous or abnormal data. The variational autoencoder is better are detecting anomalies than the autoencoder, however, even the better model can detect about 52.67 percent of the anomalies. Most of the anomalies are misclassified as normal data. This shows that both the models are having trouble with telling the difference between normal and abnormal data.

There might be 2 reasons for the underperformance of the models. First, the models lack a mechanism to detect temporal patterns. Since most of the anomalies in this data are temporal anomalies, the models are having trouble with detecting these anomalies. An LSTM deep learning classification model might be better at classifying anomalies than the models recommend in this project. An LSTM autoencoder with also be very effective at detecting. However, a conclusive study needs to be conducted comparing the models implemented and the models suggested to see which models are better at finding the results.

Second, there seem to be random distortions in the data in both training and testing samples. The readings plotted in figure 1 and figure 2 show that there are some distortions observed in both training and testing. These distortions might be interfering with the training of the data. However, this theory cannot be validated. All the samples are selected at random for the original PhysioNet/CinC Challenge[2]. These signals need be validated, and the random distortions need to be removed from the data manually. The models should be trained and tested with the new data and performance should be compared.

References

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